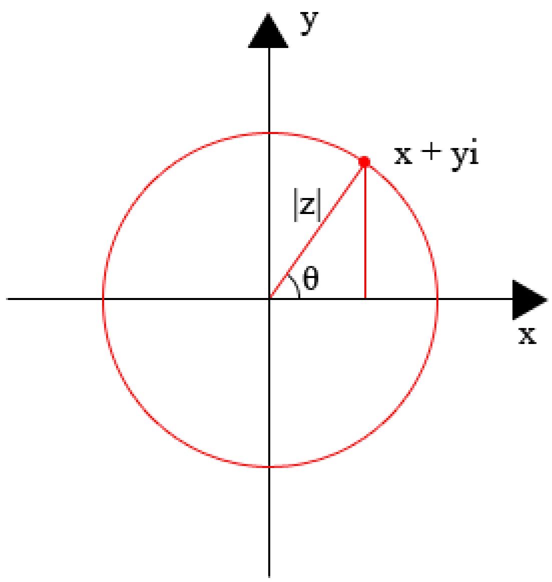
The complex plane



The polar form of a complex number

$z = r(\cos(\theta) + i*\sin(\theta))$

DeMoivre's Theorem:

if $z = r(\cos(\theta) + i*\sin(\theta))$

then $z^n = r^n(\cos(n\theta) + i*\sin(n\theta))$

Example:

$z = (1 + i) = 2^{1/2}(\cos(\pi/4)+ i*\sin(\pi/4))$

$z^{1/2} = 2^{1/4}(\cos(\pi/8) + i*\sin(\pi/8)) =$

$2^{1/4}(0.9238 + 0.3826i) = 1.0985 + 0.4549i$